

- RES.6-012 Introduction to Probability - Lectures 24, 25, 26
Textbook Notes - Sections 7.1 to 7.4

Architecture

$$\underbrace{P(X_{n+1} = j \mid X_n = i, \text{past}) = p_{ij}}_{7.1 \text{ Markov Property}} \longrightarrow \underbrace{r_{ij}(n) = \sum_k n_k(n-1) p_{kj}}_{\text{Chapman-Kolmogorov}}$$

$$\underbrace{\text{recurrent / transient, periodicity}}_{7.2 \text{ which states persist}} \xrightarrow{\text{single aperiodic class}} \underbrace{\pi_j = \sum_k \pi_k p_{kj}, \sum_k \pi_k = 1}_{7.3 \text{ long-run behaviour.}}$$

↓ start in a transient state

$$\underbrace{a_i = \sum_j p_{ij} a_j \mid \mu_i = \mathbf{I} + \sum_j p_{ij} \cdot \mu_j}_{7.4 \text{ short-run} \rightarrow \text{where and when absorbed.}}$$

- 7.1 sets up the model & n-step recursion
- 7.2 classifies states / determine whether a limit exists
- 7.3 gives long run distribution
- 7.4 handles short-run: which recurrent class entered, when.

• Section 7.1 Discrete-Time Markov Chains

• Markov Property

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = p_{ij}$$

- ↳ Next state depends on past only through current state
- ↳ Transition probabilities are time-invariant.

• Specification

- A model is given by:
 - states $S = \{1, \dots, m\}$;
 - pairs (i, j) with $p_{ij} > 0$
 - and the p_{ij} values.

- Encode model as:

- OR ↳ transition probability matrix $[p_{ij}]$ (rows sum to 1)
 - ↳ transition probability graph (nodes = states, arcs = transitions)
 - ↳ self-transitions $p_{ii} > 0$ are permitted.

• Probability of a state sequence

$$P(X_0 = i_0, \dots, X_n = i_n) = P(X_0 = i_0) p_{i_0 i_1} p_{i_1 i_2} \dots p_{i_{n-1} i_n}$$

- ↳ method: multiply the arc probabilities along the path.

• n-Step Transition Probabilities

$$r_{ij}(n) = P(X_n = j \mid X_0 = i)$$

• Chapman-Kolmogorov Equation

$$r_{ij}(n) = \sum_{k=1}^m r_{ik}(n-1) p_{kj}, \quad n > 1, \quad r_{ij}(1) = p_{ij}$$

↳ Derivation: total prob theorem on state at time $n-1$, then apply Markov Property to drop conditioning on X_0 .

↳ Matrix form: $R(n) = P^n$

↳ the n -step matrix is the n th power of the transition matrix.

• Section 7.2 Classification of States

• Accessibility

j is accessible from i if $r_{ij}(n) > 0$ for some n
↳ a path of positive probability ~~from~~ runs from i to j .
↳ Write $A(i)$ for the set of states accessible from i .

• Recurrent & Transient

• i is recurrent if every $j \in A(i)$ has $i \in A(j)$
↳ from anywhere reachable, return is possible.

• i is transient otherwise, some $j \in A(i)$ cannot return to i
↳ A transient is visited only finitely many times, $r_{ii}(n) \rightarrow 0$

• Recurrent Classes & Decomposition

For recurrent i , the set $A(i)$ is a recurrent class

↳ all its states are recurrent, mutually accessible, and no state outside is ~~acc~~ accessible from it.

↳ once entered, the chain never leaves.

• Decomposition

↳ Any Markov chain splits into one or more recurrent classes plus (possibly) some transient states.

↳ At least one recurrent class exists

↳ A recurrent state is reachable from every state accessible from it.

↳ A transient state is ^{not} reachable from ^{at least one such} ~~every~~ state ~~accessible from~~.

• Periodicity

• A recurrent class is periodic with period $d > 1$ if its states partition into d ~~subjects~~ subsets $S_1, \dots, S_d$ such that transitions from S_k always go to S_{k+1} (and from S_d to S_1).

Otherwise, aperiodic

• Test for Aperiodicity:

↳ The class R is aperiodic iff there exist $n \geq 1$ and $i \in R$ with $r_{ij}(n) > 0$ for all $j \in R$

→ a single time at which every state is reachable.

⇒ Equivalently: $r_{ij}(n) > 0$ for all $i, j \in R$ for some n .

Explanation

Two states $p_{12} = p_{21} = 1$



↳ At 1 at $0, 2, 4, \dots$ at 2 at $1, 3, 5, \dots$

↳ Return to 1 only at even times

↳ Period 2

States split into d groups arranged in a cycle.
Every transition advances you one group forward.

• Section 7.3 Steady-State Behaviour

• Steady-State Convergence Theorem

For a chain with a single recurrent class that is aperiodic, there exist π_j with :

(a) $\lim_{n \rightarrow \infty} r_{ij}(n) = \pi_j$ for all i independent of the starting state

(b) The π_j are the unique solution to the balance equations plus normalisation :

$$\pi_j = \sum_{k=1}^m \pi_k p_{kj}, \quad j = 1, \dots, m$$

$$\sum_{k=1}^m \pi_k = 1$$

(c) $\pi_j = 0$ for transient states, $\pi_j > 0$ for recurrent states

→ Both conditions are needed

• Multiple recurrent classes → the limit depends on the initial. Periodic class → $r_{ij}(n)$ oscillations and no limit exists

→ Solution method : 1. the m balance equations are linearly dependent (any one follows from the rest),
2. discard one and use the normalisation eqn in its place

• Stationary Distribution

The π_j form a probability distribution on the state space.

If $P(X_0 = j) = \pi_j$, then $P(X_n = j) = \pi_j$ for all n .

↳ the distribution is unchanged by transitions.

• Frequency Interpretations

State frequency $\rightarrow \pi_j$ is the long-run expected fraction of time spent in j .

$$\pi_j = \lim_{n \rightarrow \infty} \frac{V_j(n)}{n}$$

$\hookrightarrow$ where $V_j(n)$ is the expected number of visits to j in the first n transitions, ~~from~~ starting from i .

Transition Frequency $\rightarrow \pi_j p_{jk}$ is the long-run expected fraction of time transitions that move the state from j to k .

$\hookrightarrow$ under this reading, the balance equations, $\pi_j = \sum_k \pi_k p_{kj}$ says the frequency of entering j equals the sum of frequencies of transitions into j .

• Birth-Death Processes

A chain where transitions go only between adjacent states i and $i+1$.

Write $b_i = p_{i, i+1}$ (birth) and $d_i = p_{i+1, i}$ (death)

$\rightarrow$ Local Balance Equations

$$\pi_i b_i = \pi_{i+1} d_{i+1}, \quad i = 0, 1, \dots, m-1$$

Why? ~~the~~ every crossing of the boundary between $\{0, \dots, i\}$ and $\{i+1, \dots, m\}$ in one direction must be matched by a crossing back.

Method: solve recursively for each π_i in terms of π_0 ,
then apply normalisation.
Quicker than solve Balance Eqns.

Standard case: constant $b_i = b$, $d_i = d$, with $\rho = b/d$

$$\pi_i = \pi_0 \cdot \rho^i, \quad \pi_0 = \begin{cases} \frac{1-\rho}{1-\rho^{n+1}}, & \rho \neq 1 \\ \frac{1}{n+1}, & \rho = 1 \end{cases}$$

↳ A truncated geometric distribution.

As $n \rightarrow \infty$: if $\rho < 1$, $\pi_i = \rho^i (1-\rho)$;
if $\rho > 1$, $\pi_i \rightarrow 0$ and no steady-state
distribution exists.

$$P(\text{First observed transition birth}) = \sum_i \pi_i b_i$$

$$P(\text{first transition changes state}) = \sum_i \pi_i (b_i + d_i)$$

$$P(\text{first change of state is birth}) = \frac{P(\text{birth} \mid \text{change of state})}{\sum_i \pi_i (b_i + d_i)} = \frac{\sum_i \pi_i b_i}{\sum_i \pi_i (b_i + d_i)}$$

• Section 7.4 Absorption Probabilities & Expected Time to Absorption.

• Setup

Short-run behaviour starting from a transient state.
Since behaviour after reaching a recurrent state is ~~relevant~~ ^{irrelevant},
treat every recurrent state k as absorbing:
$$p_{kk} = 1, \quad p_{kj} = 0 \quad \text{for } j \neq k$$

Tip: To find the probability of entering a whole recurrent class, lump that class into a single absorbing state.

• Absorption Probabilities

Fix an absorbing state s .

Let $a_i = P(\text{eventually reach } s \mid X_0 = i)$.

Then:

$$\begin{cases} a_s = 1, & a_i = 0 \text{ for absorbing } i \neq s \\ a_i = \sum_{j=1}^m p_{ij} a_j & \text{for transient } i. \end{cases}$$

→ Derivation: total prob theorem on first transition, then the Markov property.

→ Gambler's Ruin: Fortune i , target m , win probability p each round, $p = (1-p)/p$
→ The probability of reaching m before 0:

$$a_i = \begin{cases} \frac{1-p^2}{1-p^m}, & p \neq 1 \\ \frac{i}{m}, & p = 1 \end{cases}$$

↳ For $p < 1/2$ ($p > 1$), $a_i \rightarrow 0$ as $m \rightarrow \infty$ for any fixed i .

• Expected Time to Absorption

$$\mu_i = E[\text{transitions until absorption} | X_0 = i] :$$

$$\left\{ \begin{array}{l} \mu_i = 0 \text{ for recurrent } i, \\ \mu_i = 1 + \sum_{j=1}^m P_{ij} \mu_j \text{ for transient } i \end{array} \right\}$$

→ Derivation: time from i is 1 step plus the expected time from the next state j apply total expectation

• Mean First Passage & Recurrent Times

For a chain with a single recurrent class, fix a recurrent state s .

$$\text{Mean first passage time } t_i = E[\text{steps to reach } s | X_0 = i] :$$

$$t_s = 0, \quad t_i = 1 + \sum_{j=1}^m P_{ij} \cdot t_j \text{ for } i \neq s$$

Method: convert s to an absorbing state, making all others transient, then apply the expected-time-to-absorption equations.

$$\text{Mean recurrence time: } t_s^* = E[\text{steps to first return to } s | X_0 = s]$$

$$t_s^* = 1 + \sum_{j=1}^m P_{sj} \cdot t_j$$

Computed once the t_i are known.

Note the difference: $t_s = 0$ by definition, whereas t_s^* counts a genuine return, requiring at least one step.